

Customer Behavior Analysis Report

Introduction

This project focuses on analyzing customer behavior using purchase data. The objective of the analysis is to understand customer purchasing patterns, payment preferences, customer segmentation, and churn behavior. The analysis helps businesses make better marketing and customer retention decisions.

Dataset Overview

The dataset contains customer purchase information including:

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity
- Total Purchase Amount
- Payment Method
- Customer Age
- Gender
- Churn

The dataset was analyzed using Python libraries such as Pandas and Matplotlib in Google Colab.

Data Cleaning

The following data cleaning steps were performed:

* Removed duplicate records

- * Handled missing values
- * Checked dataset structure and column information
- * Verified data consistency

These steps improved the quality of the dataset for further analysis.

Feature Engineering

New features were created to improve the analysis:

Average Purchase Value

Calculated by dividing the total purchase amount by quantity purchased.

Age Group

Customers were categorized into:

- * Teen
- * Young Adult
- * Adult
- * Middle Age
- * Senior

Customer Segment

Customers were segmented into:

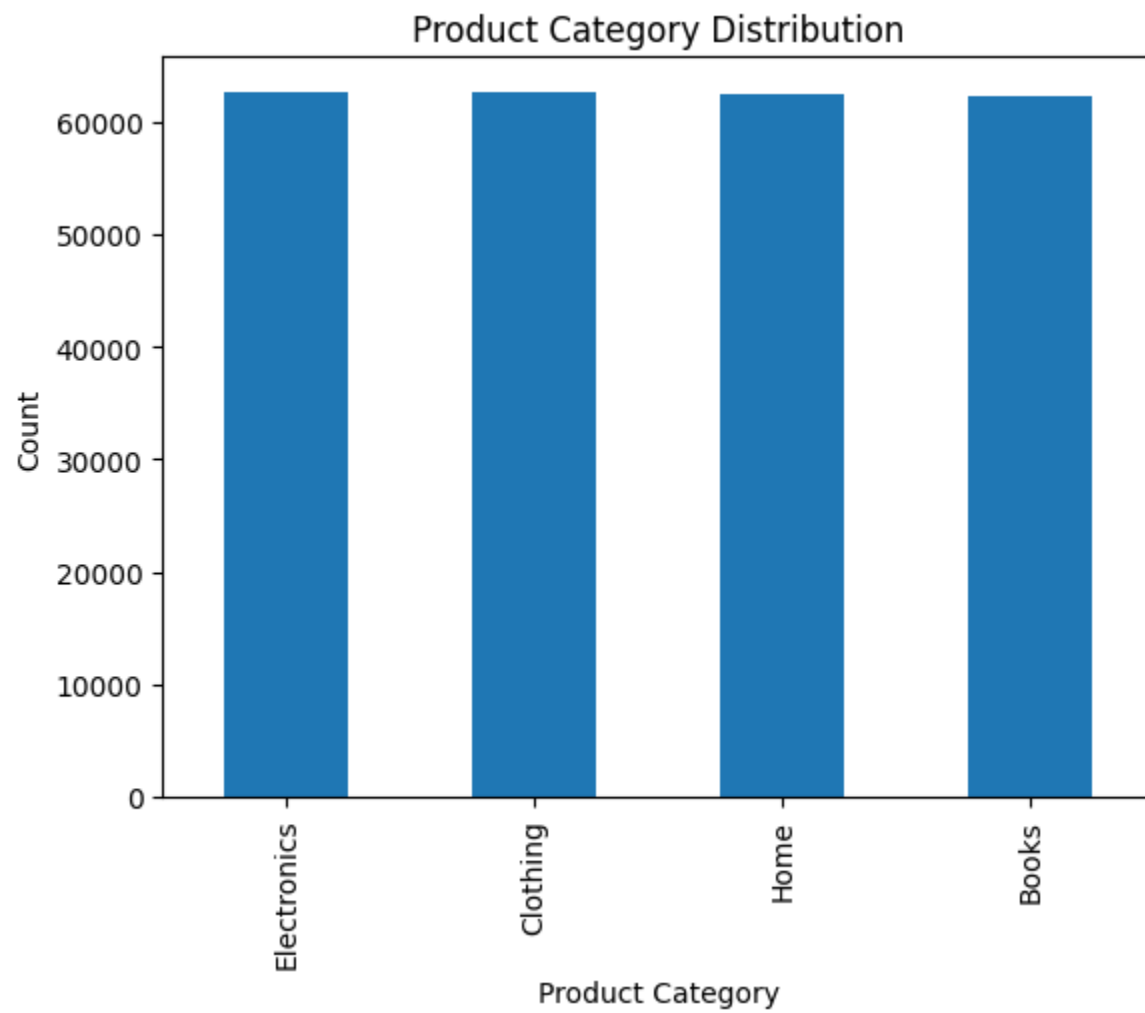
- * Low Value
- * Medium Value
- * High Value

based on their total purchase amount.

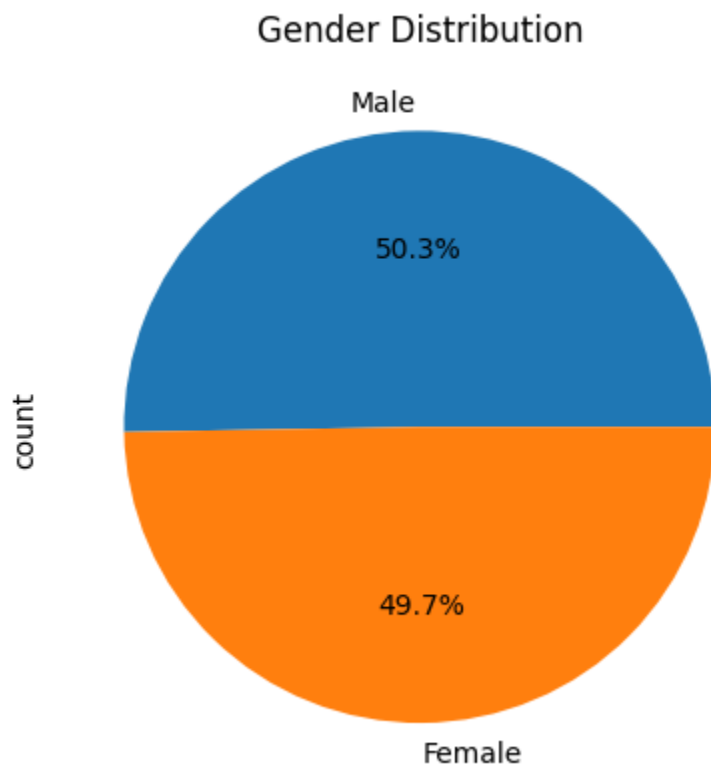
Visualizations

The following visualizations were created:

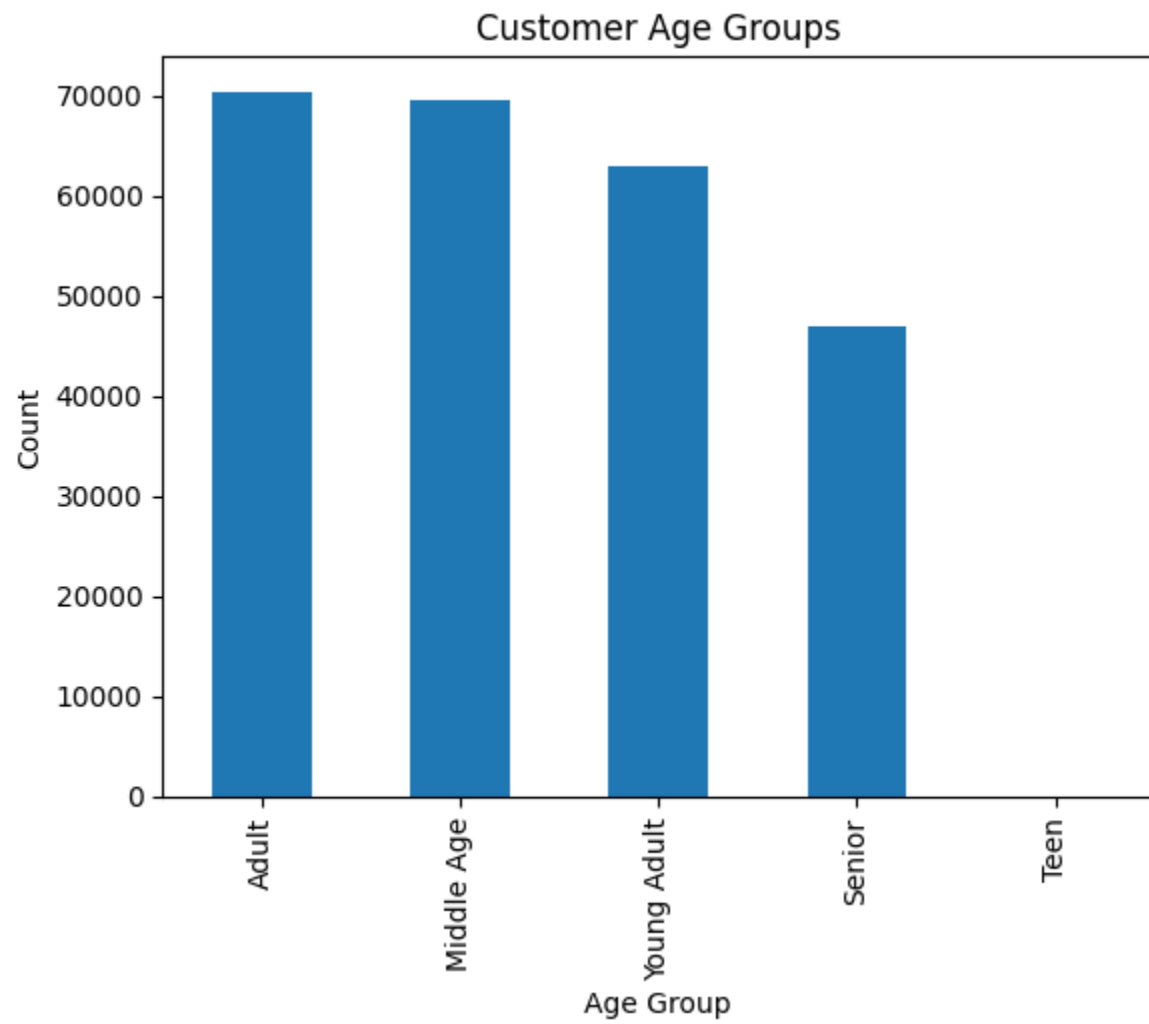
* Product Category Distribution



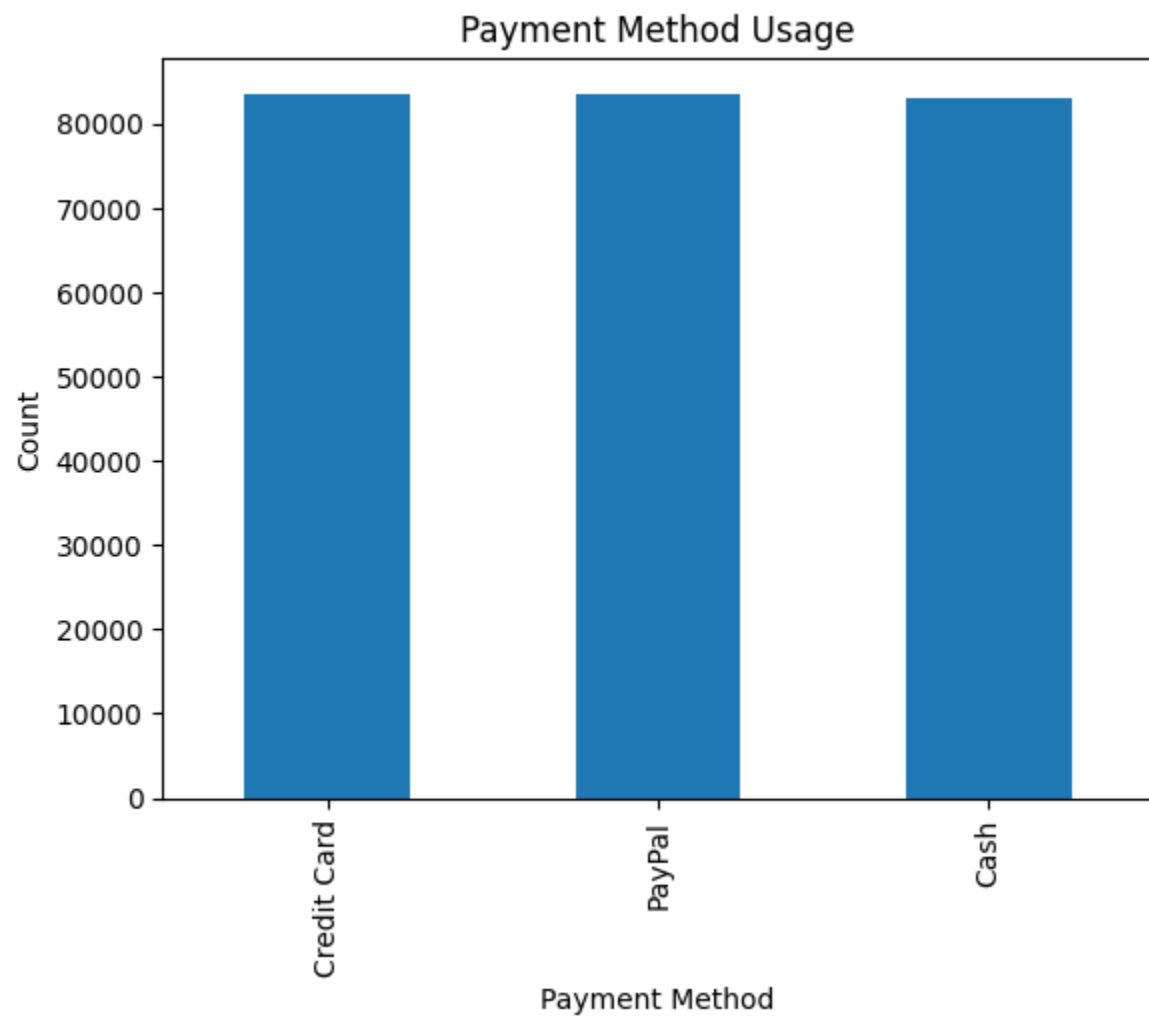
* Gender Distribution



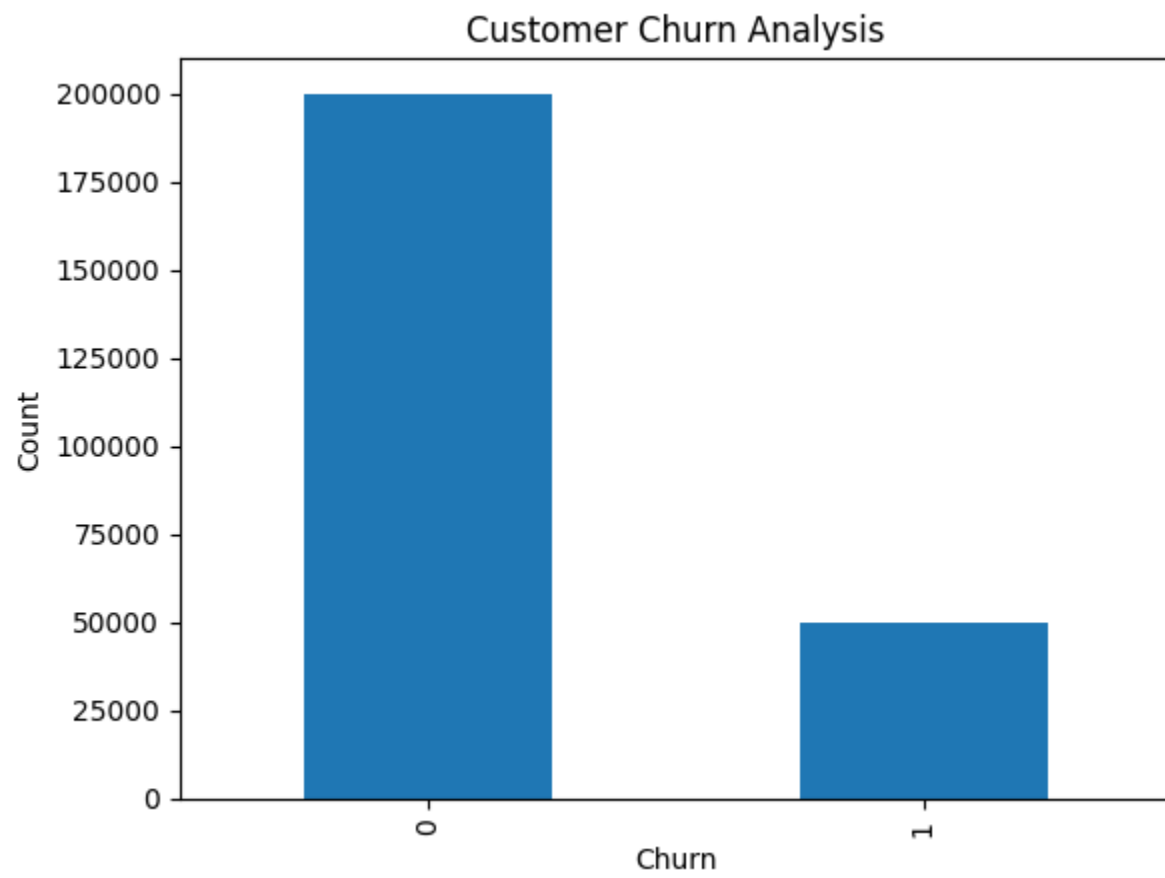
* Age Group Distribution



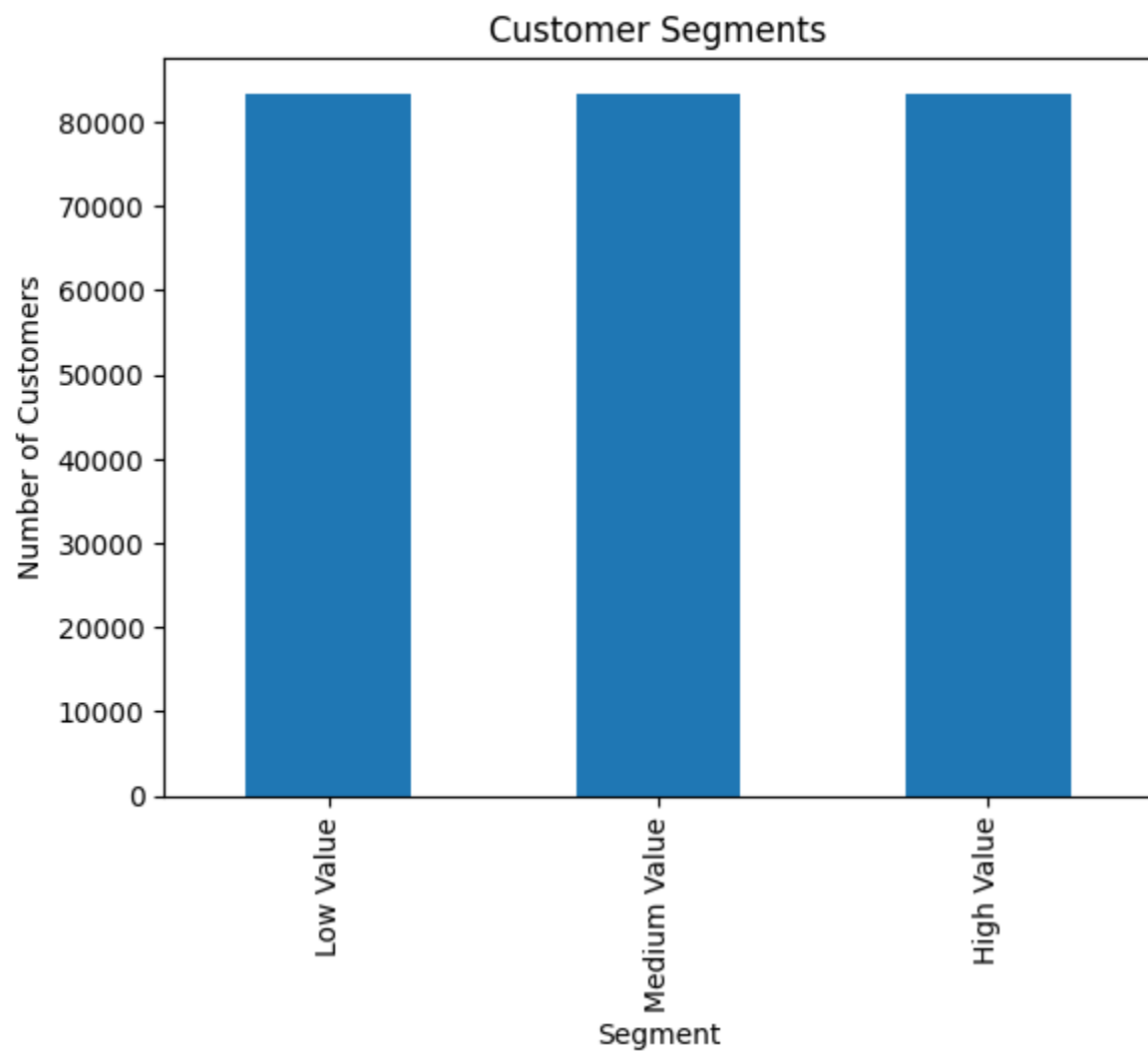
* Payment Method Usage



* Customer Churn Analysis



* Customer Segmentation



* Total Purchase Amount by Category



Customer Segmentation

Customers were grouped into three categories based on their purchase amount.

Low Value Customers

These customers spend less and purchase fewer products. Promotional offers can help improve their engagement.

Medium Value Customers

These customers contribute moderate revenue and have the potential to become high-value customers.

High Value Customers

These customers contribute the highest revenue to the business. Retention strategies and loyalty programs are important for this group.

Findings / Observations

- * Medium-value customers form a large portion of the customer base.
- * Certain product categories generate higher sales revenue.
- * Digital payment methods are widely used by customers.
- * Young adults contribute significantly to purchases.
- * Customer churn exists and affects customer retention.

Recommendations

1. Focus on retaining high-value customers through loyalty programs.
2. Provide personalized offers for low-value customers.
3. Increase promotions for popular product categories.
4. Improve customer retention strategies to reduce churn.
5. Use targeted marketing based on age groups and purchase behavior.

Conclusion

The customer behavior analysis provided valuable insights into customer purchasing patterns, segmentation, and churn behavior. The analysis can help businesses improve marketing strategies, customer engagement, and overall business performance.